

A ten-vertex simplicial tree with 11-torsion in its Laplacian group

Disproof of Conjecture 00000004215

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Abstract

We give a simplicial two-dimensional spanning tree on 10 vertices whose integer up-down Laplacian has a nonzero cokernel element of order 11. The same element belongs to the first critical group. Thus a Laplacian torsion prime can exceed the number of vertices. All boundary and torsion certificates are exact integers; the Lean project checks them and proves the tree conditions and quotient-group conclusion. Exact Python verification and discovery code are included.

The complex and conventions

Let X have vertex set $\{0, \dots, 9\}$, every one of its 45 edges, and the following 36 triangles, with no higher-dimensional faces:

156	128	357	013	368	179
018	149	017	024	148	358
256	258	468	034	478	236
249	239	039	345	012	047
456	459	367	057	126	059
679	068	069	278	267	089

Here abc denotes $\{a, b, c\}$, oriented with $a < b < c$. All listed triangles are distinct. Every edge occurs in a listed triangle. This defines a pure simplicial complex; it is a subcomplex of the complete two-skeleton of the simplex on these ten vertices.

Order edges lexicographically and triangles in the displayed order, reading each row from left to right. Let

$$D : \mathbb{Z}^{45} \longrightarrow \mathbb{Z}^{10}, \quad A : \mathbb{Z}^{36} \longrightarrow \mathbb{Z}^{45}$$

be the oriented boundary matrices: $De_{ab} = e_b - e_a$ and $Ae_{abc} = e_{bc} - e_{ac} + e_{ab}$. We use the standard up-down Laplacian $L = AA^\top$ on edges, as in Duval–Klivans–Martin, Section 3. Its integer cokernel is $\mathbb{Z}^{45}/L\mathbb{Z}^{45}$, and the first critical group is $K_1(X) = \ker D/\operatorname{im} L$. This convention is explicit: the total Hodge Laplacian includes a further term $D^\top D$.

Verification that this is a simplicial tree

The first nine edges are $01, 02, \dots, 09$. Delete their rows from A to obtain a square matrix B of size 36. Let E be the 45×36 matrix whose column indexed by $1 \leq a < b \leq 9$ is

$$E_{ab} = e_{ab} - e_{0b} + e_{0a}.$$

These are the fundamental cycles for the star with center 0. Directly from the boundary formulas, $A = EB$, and the bottom 36 rows of E form the identity matrix.

Exact integer elimination gives $\det B = -11$. The supplied `verify.py` implements fraction-free Bareiss elimination and also independently calculates $C = 11B^{-1}$ using rational Gaussian elimination. Every entry of C is an integer. Set

$$H = \begin{bmatrix} 0_{36 \times 9} & C \end{bmatrix}, \quad Se_0 = 0, \quad Se_j = 11e_{0j} \quad (1 \leq j \leq 9).$$

The full integer coefficient table for H appears in `lean/Main/Certificate.lean`. It supplies the following directly checkable certificates:

$$DA = 0, \quad HA = 11I_{36}, \quad AH + SD = 11I_{45}. \quad (1)$$

Both Lean's kernel and the independent exact Python verifier check these identities from the listed triangles. The formal proof uses (1) directly, so it does not assume a determinant computed by an external program.

If $Az = 0$ for an integer 2-chain z , then $11z = HAz = 0$ and hence $z = 0$. There are no 3-simplices, so $H_2(X; \mathbb{Z}) = 0$. If $Dz = 0$ for a rational 1-chain z , then

$$z = A \left(\frac{1}{11} Hz \right).$$

Thus $H_1(X; \mathbb{Q}) = 0$. In fact every integer cycle has $11z = AHZ$. The complete one-skeleton is connected, and

$$f_2 = 36 = 45 - 10 + 1 = f_1 - f_0 + 1.$$

These are the usual spanning-tree conditions for a two-dimensional subcomplex containing the full one-skeleton. In particular X is a simplicial spanning tree of the complete two-skeleton on ten vertices. The definition permits finite integral H_1 ; requiring integral acyclicity would exclude precisely the torsion phenomenon under discussion.

A nonzero element killed by 11

Set

$$c = e_{01} - e_{04} + e_{14}, \quad b = 11c.$$

Clearly $Dc = 0$. Define an integer edge vector w to be zero on $01, \dots, 09$; on the remaining edges, in lexicographic order, its 36 coordinates are the entries below, read row by row:

$$\begin{array}{cccccc} 33 & 0 & 371 & 98 & 53 & 44 \\ 44 & 188 & -7 & 33 & 43 & 20 \\ 63 & -22 & -73 & 33 & -51 & -6 \\ -29 & -72 & -33 & -117 & -184 & -55 \\ -250 & -139 & -45 & 11 & -43 & -11 \\ -12 & -44 & 44 & -140 & 100 & 0 \end{array}$$

Direct integer multiplication with the boundary matrix of the listed triangles gives

$$Lw = 121c = 11b, \quad w^\top L = 121c^\top, \quad w^\top b = 4081 = 121 \cdot 33 + 88. \quad (2)$$

The second identity also follows by transposing the first, because L is symmetric. The last uses $w_{01} = w_{04} = 0$ and $w_{14} = 371$. These identities are checked in both supplied verification systems; no numerical approximation is involved.

Theorem 1. *The class of b in $\mathbb{Z}^{45}/L\mathbb{Z}^{45}$ is nonzero and has order 11. Its class in $K_1(X)$ has the same property.*

Proof. The first identity of (2) implies $11[b] = 0$. If $b = Lz$ for an integer vector z , then

$$4081 = w^\top b = w^\top Lz = 121c^\top z,$$

which is impossible modulo 121. Hence $[b] \neq 0$. Since 11 is prime, its order is exactly 11. Also $Db = 0$, so the same nonzero class and annihilation identity hold in $\ker D/\text{im } L$. $\square$

The Smith normal form of an integer matrix presents its cokernel. The exhibited element therefore certifies 11-torsion in the Laplacian cokernel. Since $11 > 10$, the conjectured vertex bound is false. No assertion about which lens-space examples attain a corrected bound is needed to refute the stated universal bound.

Formalization and reproduction

The Lean project constructs the actual oriented edge and triangle boundaries from finite vertex tuples. It proves that the edge list is precisely the complete one-skeleton, that triangles are valid and distinct, that the associated face family is downward closed, and that it has 36 facets. The tree predicate records the explicit chain conditions: integral top-cycle vanishing and rational first-cycle exactness. These are proved using the integer certificates (1), with rational extension performed algebraically in Lean.

The critical group is an actual additive quotient: `cycles` is the kernel of the edge boundary, and `lapImage` is the range of the Laplacian into that kernel. `torsion_nonzero` and `torsion_eleven` prove the two properties of the exhibited class. A separate theorem, `cokernel_torsion`, proves them in the full matrix cokernel. The final `counterexample` theorem combines the tree conditions, primality of 11, the comparison $11 > 10$, and the nonzero class killed by 11.

`Certificate.lean` contains explicit integers and kernel-checked decisions. The initial numerical search is used only to find a candidate; it is not a proof oracle. The `discovery.py` script preserves that basis-exchange search (requiring NumPy), including an exact determinant check of its output. The proof and `verify.py` require no NumPy: the latter uses only Python’s standard library, checks all relevant integer identities, and checks its data against the Lean source.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The fresh build checks the entire certificate module. There are no proof placeholders, custom axioms or external computation oracles. Run `python verify.py` from the submission directory for the independent arithmetic verification, and `tectonic main.tex` to regenerate the PDF. The author performed an adversarial self-review; no independent reviewer or subagent was used.

References

1. The Last Math Competition, Conjecture 00000004215. The exact bilingual source and its upstream provenance are included.
2. A. M. Duval, C. J. Klivans and J. L. Martin, *Critical Groups of Simplicial Complexes*, Section 3, Definitions 3.1–3.2. Authors’ preprint (2011), available here.